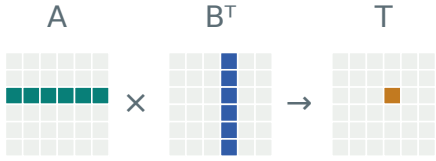


TriMul: contract globally, reuse locally

A channel-wise contraction connects two fused endpoints. Training adds explicit saved tensors and a separate backward pipeline.

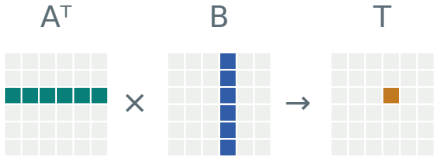
A / THE MATHEMATICS

Outgoing

$$T_{ijc} = \sum_k A_{ikc} B_{jkc}$$


Sum over k; one pair matrix for each channel c.

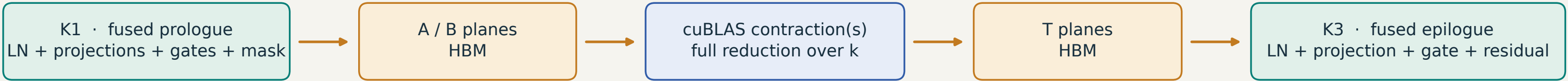
Incoming

$$T_{ijc} = \sum_k A_{kic} B_{kjc}$$


The contracted axis changes; the model semantics stay explicit.

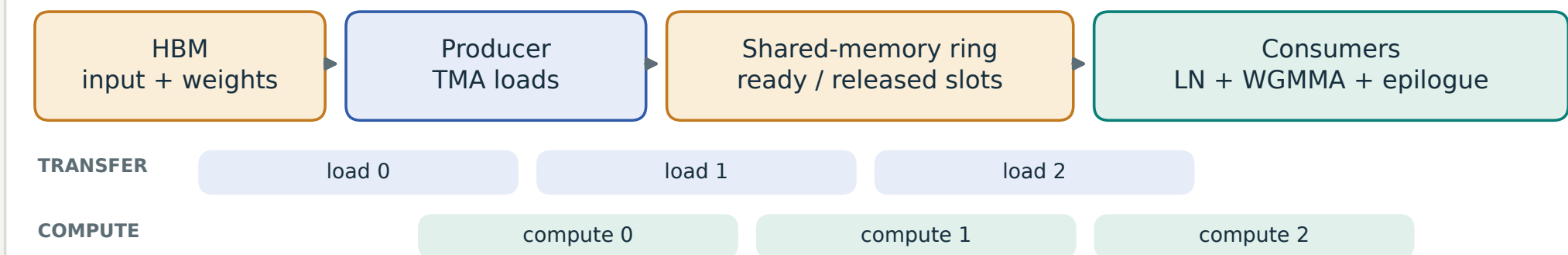
Bidirectional path
separate channel groups
for the two directions

B / THE EXECUTION BOUNDARIES



These HBM intermediates are retained in the packaged inference path. Kernel fusion does not remove the contraction boundary.

INSIDE A K1 / K3 CTA · schematic, not a timing trace



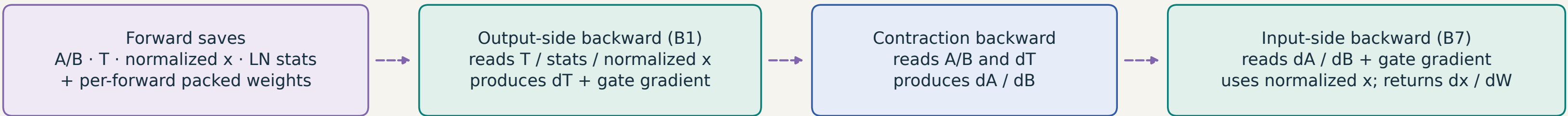
Slot depth and consumer count vary by config.

WHY STOP FUSING HERE?

- Contraction consumes tiles across the k axis.
- CTA-local storage is finite.
- More fusion can raise register / smem pressure.
- A different boundary requires a different algorithm.

The diagram shows current boundaries, not an impossibility proof.

C / TRAINING TENSOR LIFETIMES · D128 PACKAGED PATH

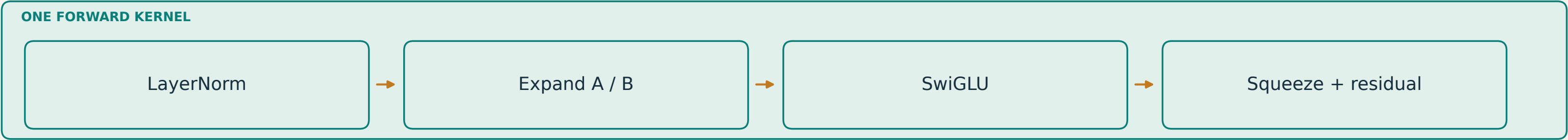


Saved buffers live across forward and backward; gradients and partial reductions may add launches. D256+ research checkpoints are separate.

Transition: keep the expanded activation local

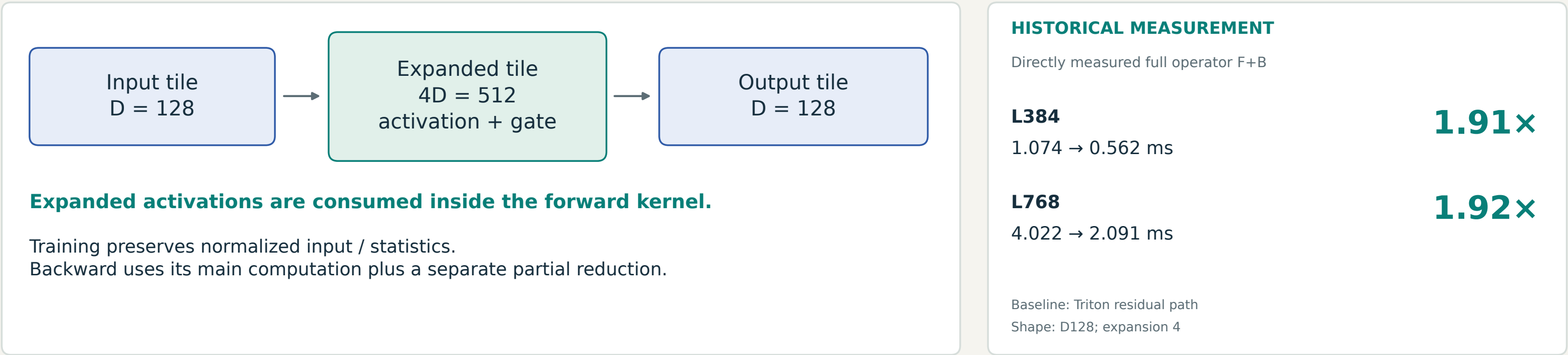
D128 · expansion 4 · BF16 · SM90 specialized path [C4]

A / EXECUTION AND FUSION



The enclosing boundary is one kernel; the inner boxes are mathematical stages.

B / ALGORITHM AND DATA REUSE

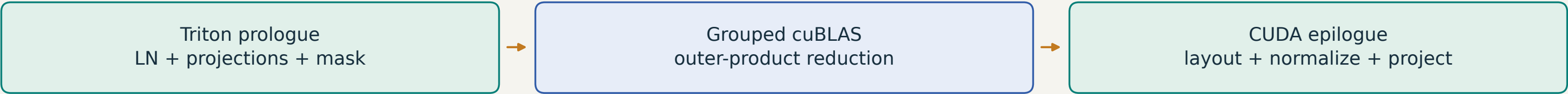


The fusion boundary applies to this specialized shape. Other shapes can follow different dispatch paths.

Outer Product Mean: fuse around the sequence reduction

MSA → pair representation · training path [C5]

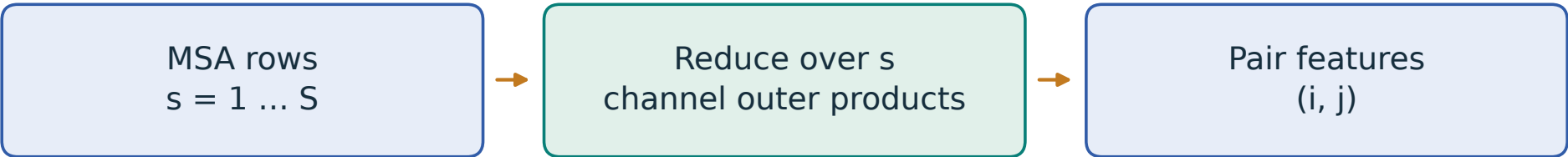
A / EXECUTION AND FUSION



A/B and the product cross global-memory boundaries. Mask-count normalization is computed separately.

B / ALGORITHM AND DATA REUSE

$$O_{ijab} = \sum_s A_{sia} B_{sjb}$$



Backward choice: save O, or recompute O to reduce stored activations.

The performance card uses save O; both variants remain in the data bundle.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384

5.858 → 2.028 ms

2.89×

Baseline: Own engine path (job 15640)

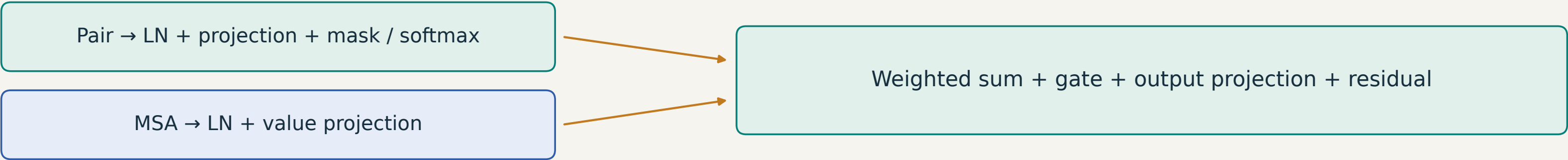
Shape: S1024; save O

The sequence reduction is retained as a library call; epilogue fusion combines layout, normalization and projection.

Pair-Weighted Averaging: join two reusable input branches

Independent pair-weight and MSA-value paths feed a fused output computation [C6]

A / EXECUTION AND FUSION



B / ALGORITHM AND DATA REUSE

PAIR WEIGHTS

$$P_{ijh} = \text{softmax}_j(b_{ijh})$$

$$U_{sihd} = \sum_j P_{ijh} V_{sjhd}$$

Pair weights are reused across MSA rows **s**.

Gate and output projection consume the weighted result in the fused path.

MSA VALUES

$$V_{sjhd}$$

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384

4.161 → 1.561 ms

2.67x

Baseline: Own engine path (job 15737)

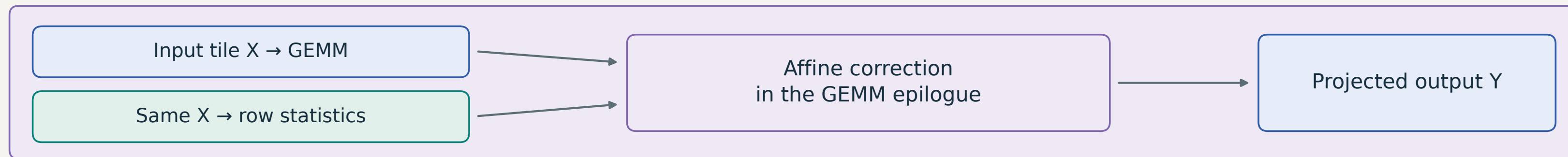
Shape: S1024; D_msa64 / D_pair128

Arrows show independent inputs. Training mask preparation and multi-kernel backward are outside the forward boxes.

LayerNormLinear: normalize through the GEMM epilogue

CuTe SM90 main-kernel algorithm [C7] · weight preparation is separate

A / EXECUTION AND FUSION



One main kernel. No full normalized activation is materialized in HBM.

B / ALGORITHM AND DATA REUSE

ALGEBRAIC REARRANGEMENT

$$Y = r[X(\gamma W) - \mu(\mathbf{1}^T \gamma W)] + \beta W$$

GEMM and row statistics use the same input tile.
 Mean / reciprocal standard deviation correct the output.
 Weight-side terms are prepared outside the main kernel.

The diagram describes CuTe; the portable timing is a different path.

HISTORICAL MEASUREMENT

Portable Triton reference ONLY; not the CuTe path

L384

0.350 → 0.169 ms

2.07x

Baseline: Pre-change portable Triton

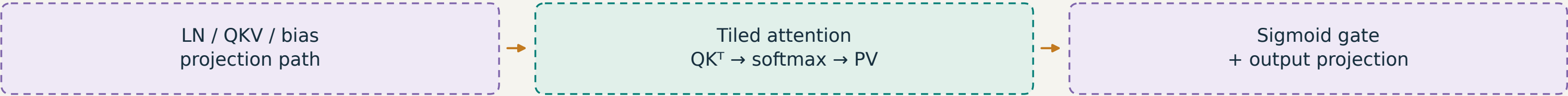
Shape: D128 to 16

The portable reference does not quantify the CUDA/CuTe fusion shown here. A matched CuTe comparison remains pending.

TriangleAttention: tiled attention between projection stages

Dispatch-dependent module [C8, C9] · dashed boundaries denote stages, not single launches

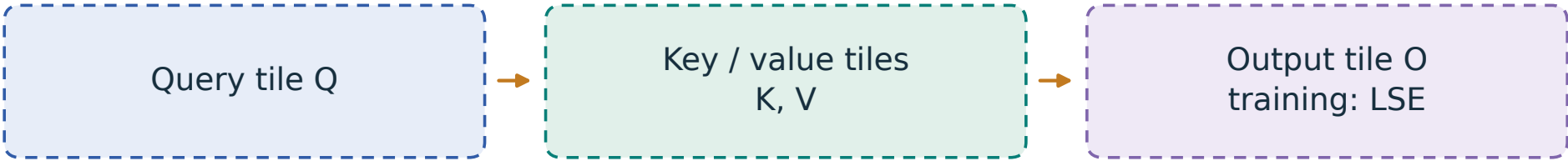
A / EXECUTION AND FUSION



Actual projection and gated-output fusion depend on shape, dtype, backend and available native artifacts.

B / ALGORITHM AND DATA REUSE

$$O = \text{softmax}(QK^T/\sqrt{d} + B) V$$



LSE supports the training path; backward has its own kernel boundaries.

Schematic only: tile schedule and exact saved tensors depend on dispatch.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

Matched comparison pending

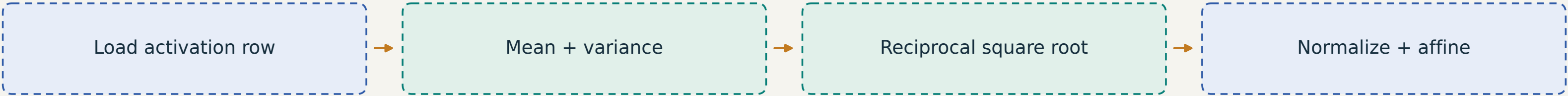
No speedup assigned to this schematic.

A full-module paired measurement is required before attaching a speedup to this figure.

LayerNorm: reduce a row, then reuse its statistics

Standalone normalization · portable Triton historical comparison [N0, N1]

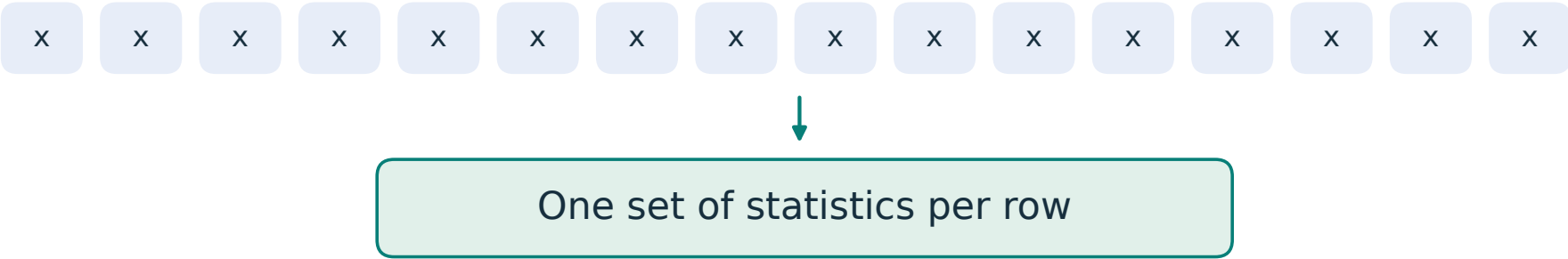
A / EXECUTION AND FUSION



Boxes show the row algorithm. Backward parameter reductions can require additional kernels.

B / ALGORITHM AND DATA REUSE

$$y_i = (x_i - \mu) (\sigma^2 + \varepsilon)^{-1/2} \gamma_i + \beta_i$$



Centering and scaling share row statistics.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384
0.276 → 0.246 ms **1.12×**

L768
1.035 → 0.949 ms **1.09×**

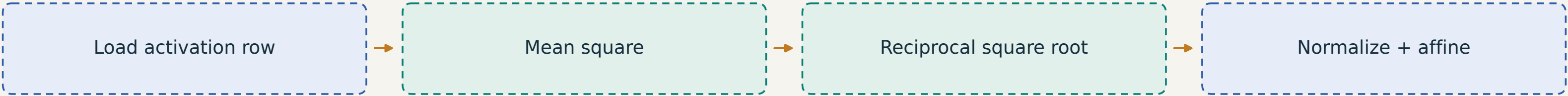
Baseline: Pre-change portable Triton
Shape: D384

Measured gains concern the standalone portable path; they do not establish cross-operator CUDA fusion speedups.

RMSNorm: reduce a row, then reuse its statistics

Standalone normalization · portable Triton historical comparison [N0, N1]

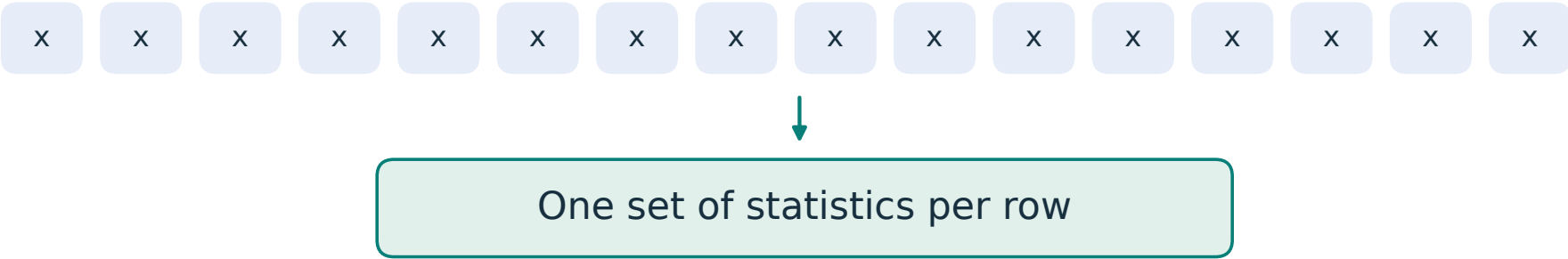
A / EXECUTION AND FUSION



Boxes show the row algorithm. Backward parameter reductions can require additional kernels.

B / ALGORITHM AND DATA REUSE

$$y_i = x_i (\text{mean}(x^2) + \varepsilon)^{-1/2} \gamma_i$$



No centering / mean subtraction.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384

0.075 → 0.072 ms

1.04×

L768

0.281 → 0.259 ms

1.08×

Baseline: Pre-change portable Triton
Shape: D128

Measured gains concern the standalone portable path; they do not establish cross-operator CUDA fusion speedups.